

Fintech Analytics Project – Mutual Fund Analytics Platform

Executive Summary

This project analyzes mutual fund industry data from 2022–2026 covering NAV performance, AUM growth, SIP inflows, investor behavior, and portfolio allocation. The objective was to build an end-to-end analytics platform including ETL pipelines, database design, exploratory analysis, performance analytics, advanced risk metrics, and an interactive Power BI dashboard.

Objectives

- Build automated data ingestion pipelines
 - Clean and validate mutual fund datasets
 - Design SQLite analytical database
 - Perform exploratory data analysis
 - Compute fund performance metrics
 - Build investor and market analytics dashboards
 - Generate advanced risk and portfolio insights
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Dataset Overview

Datasets Used:

1. Fund Master
2. NAV History
3. AUM by Fund House
4. SIP Inflows
5. Category Inflows
6. Industry Folios
7. Scheme Performance
8. Investor Transactions
9. Portfolio Holdings
10. Benchmark Indices

Total NAV Records: 46,000+

Total Investor Transactions: 32,778+

Total Schemes Analyzed: 40

Data Engineering

ETL Pipeline

- Imported all CSV datasets
- Validated schema and datatypes
- Removed duplicates
- Standardized date formats
- Generated cleaned datasets

Database Design

SQLite star schema created with:

- dim_fund
 - dim_date
 - fact_nav
 - fact_transactions
 - fact_performance
 - fact_aum
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Exploratory Data Analysis

Key Findings

1. SBI Mutual Fund maintained industry-leading AUM.
 2. SIP inflows reached record highs in 2025.
 3. Equity folios doubled between 2022 and 2025.
 4. Large-cap and flexi-cap categories attracted the highest inflows.
 5. Investor participation expanded strongly in B30 cities.
 6. Younger investors showed higher SIP participation.
 7. Maharashtra generated the highest transaction volume.
 8. Equity-oriented funds dominated industry assets.
 9. Most large-cap funds exhibited strong correlation with benchmark indices.
 10. Sector allocation remained concentrated in financial services and technology.
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Fund Performance Analytics

Metrics Computed:

- CAGR
- Daily Returns
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown
- Tracking Error

Top Sharpe Ratio Funds:

- ICICI Prudential Liquid Fund
- Mirae Asset Large Cap Fund
- Kotak Flexicap Fund
- SBI Bluechip Fund

Advanced Analytics

VaR & CVaR

95% Historical VaR and Conditional VaR calculated for all schemes.

Rolling Sharpe Analysis

90-day rolling Sharpe ratio analyzed for key funds.

Cohort Analytics

Investor cohorts grouped by transaction year and investment behavior.

SIP Continuity Analysis

Investors with SIP gaps above 35 days were flagged as at-risk.

Fund Recommendation Engine

Rule-based recommender created using risk grade and Sharpe ratio rankings.

Dashboard Overview

Page 1 – Industry Overview

- AUM KPIs

- SIP KPIs
- Folio KPIs
- AMC comparison

Page 2 – Fund Performance

- Return vs Risk analysis
- Benchmark comparison
- Fund scorecards

Page 3 – Investor Analytics

- State-wise investments
- Age group analysis
- Transaction trends

Page 4 – SIP & Market Trends

- SIP growth trends
- Category inflows
- Market comparison

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Plotly
- SQLite
- SQLAlchemy
- Power BI
- GitHub

Conclusion

The project successfully delivered a production-style mutual fund analytics solution integrating data engineering, analytics, visualization, and dashboarding. The platform enables fund performance evaluation, investor behavior analysis, and risk assessment for informed investment decision-making.

